Part A: Plan a delivery with A*

a) State: (row, column) tuple
Initial State: (row, column) relative to --start
Actions: UP, DOWN, LEFT, RIGHT
Transition Model: The agent can move and take described actions provided a wall does not prevent it from taking the action. (1,2) [UP] resulting in (1,3).
Goal Test: Current State == Goal State, (5,6) == (5,6) [True]
Path Cost: The sum of steps taken. Each step adds 1 to cost.

Because of our movement cost, the unit cost, is 1 per step. The BFS search explores all states one step away from the current state before moving on to the states that are further away. This means that the first time that the goal check returns true is also the most optimal path.

b) The Manhattan Distance does not overestimate the distance to goal, making it a safe choice for a heuristic.

Map	Delivery Task	BFS Route Length	A* Route Length	BFS Expanded Node Count	A* Expanded Node Count
Baseline	S -> W	18	18	67	22
Baseline	L -> D	22	22	79	62
Baseline	H -> P	16	16	41	16
Detour	S -> W	26	26	70	51
Detour	L -> D	22	22	65	60
Detour	H -> P	16	16	41	16

The algorithms will find routes of equal length because of BFS nature, to find the most optimal path. Although A* does not always expand fewer nodes, it also finds the most optimal route. When comparing them, it appears that A* is more efficient, but that efficiency is not guaranteed.

Part B: Learn a delivery policy

a)

S->W

In all three training sessions, for task SW, it takes some time for the agent to find its footing, regularly hitting its' step limit. In all three sessions the agent has a rudimentary understanding, and fails significantly less, around approximately the 30th episode. However it is still not especially comfortable in the environment, this is evident by high step counts. In all three sessions, comfort is established approximately between the 80th-90th episode. It is interesting to note that even in some late episodes, it stumbles and takes noticeably more steps than than preceding episodes

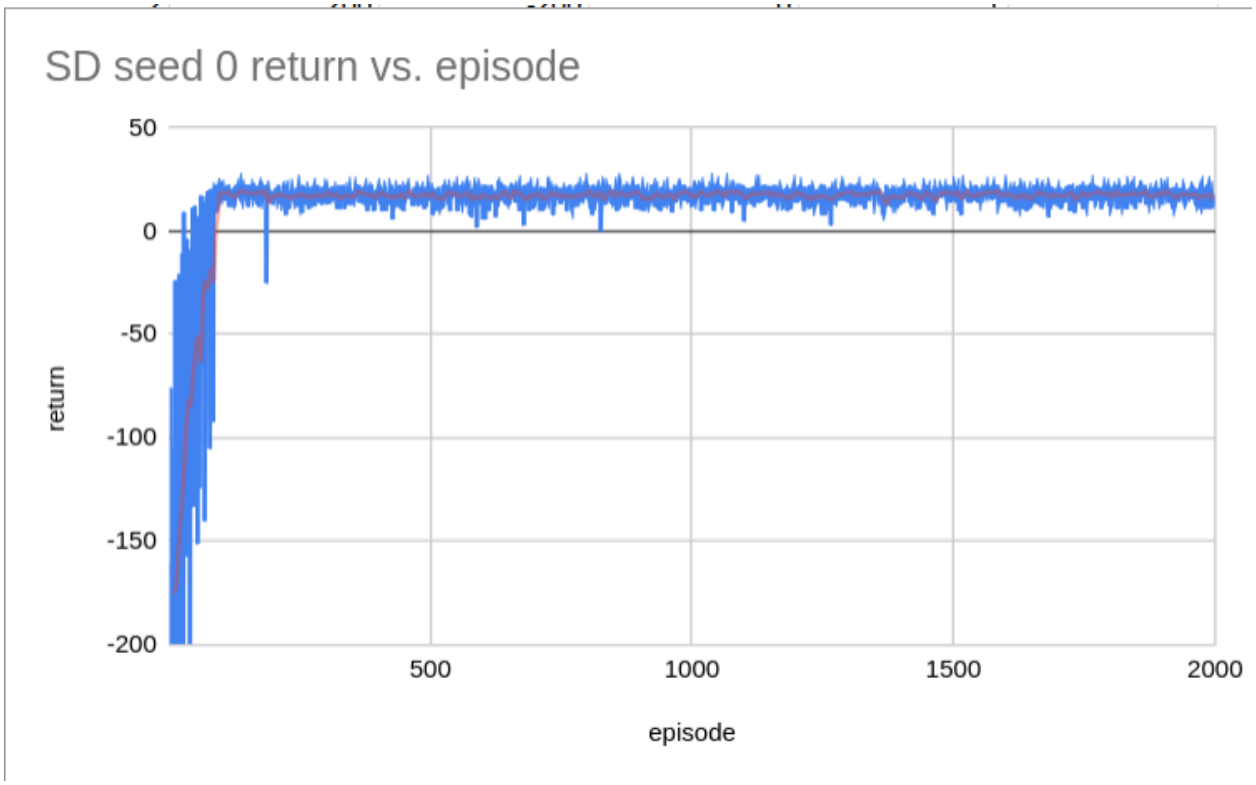
L->D

Like the previous task, the beginning episodes are noisy, chaotic and prone to failure. This task differs in that the agent picks up the route rather quickly, although not optimally at first. Around approximately the 13th-15th episode failure reduces dramatically across all three training sessions. This task seems to take longer to train, the agent does not consistently perform efficiently until approximately the 125th episode, this holds for all three training sessions. However when compared to the last task, the algorithm does not seem to be stumbling into inefficient paths. I believe this is happening because the L->D path is more readily seen as direct.

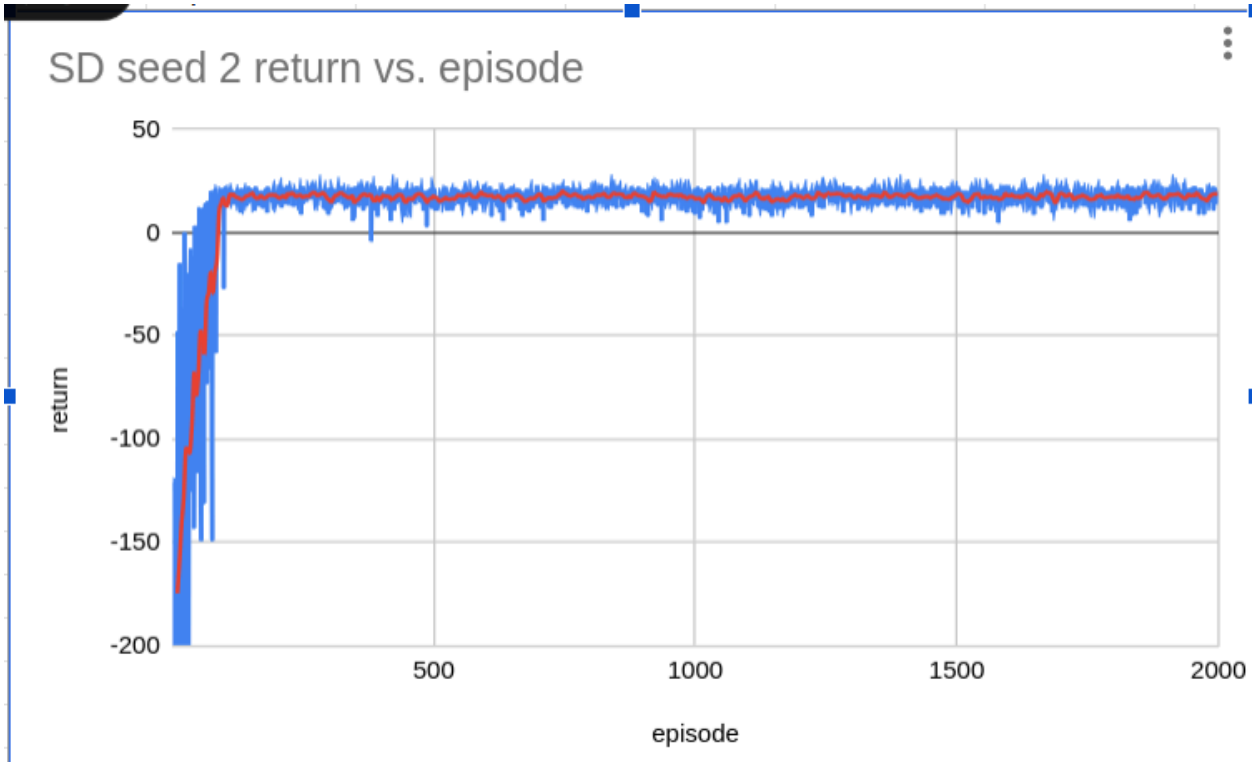
H->P

Failure reduces much sooner,and completely drops during approximately the 30th-40th episode, across all three training sessions. When visually inspecting the map it makes intuitive sense, there is really only one path, where the path could branch is not worth examining. This is evidenced by the consistency of the step count and return value, so quickly after establishing an optimal behavior.

b)


SD seed 0 return vs. episode
50
0
-50
-100
-150
-200
return
500
1000
1500
2000
episode


SD seed 1 return vs. episode
50
0
-50
-100
-150
-200
return
500
1000
1500
2000
episode



Part C: Design your own map

b)

Factor	Question	Answer	Settings
ϵ	How does exploration affect learning and final performance?	An interesting occurrence is happening, likely the result of the map design and task pair destinations. Mean return and Mean steps, 22 and 18 respectively, appear in all 3 evaluations.	.05, .2, .5

c)
(i) How fast does changing the exploration rate affect how fast the agent learns the map and most optimal route?
(ii) The map conforms to the same size as the provided campus map. There is a potential loop located in the left hand side of the map, because there are multiple wall openings near each other. This could cause a potential loop that the agent does not exit.
(iii) I'm only expanding on what was done in part B here.
(v) I had somewhat expected the agent to maybe find its way in loop and fail more regularly when challenged with a low exploration rate. However it managed to overcome that pitfall given the large training window. Even under a more stressful learning regimen it managed to efficiently establish an optimal path, as all three evaluations produced the same mean steps and returns. This was surprising, however after re-examining the map design alongside the data, it became more clear.